

VIOOLSDRIF, South Africa

WMO: 684110

Lat: 28.769S

Lon: 17.623E

Elev: 571

StdP: 14.40

Time Zone: 2.00 (E02)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	48.2	50.3	11.2	10.0	75.6	14.8	11.9	77.7	16.0	68.3	14.1	67.0	2.5	320	0.351

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	22.9	106.7	66.9	104.0	66.4	101.5	65.9	74.9	89.4	72.4	90.7	70.5	91.6	6.1	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.0	112.6	87.1	66.0	98.0	84.0	63.0	87.9	82.1	38.8	89.4	36.4	91.2	34.7	91.4	82.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.7	15.0	13.5	DB	43.0	113.4	3.7	2.7	40.3	115.4	38.2	117.0	36.1	118.5	33.4	120.5	(4)
(5)			WB	36.9	73.9	2.4	1.3	35.2	74.8	33.8	75.5	32.4	76.3	30.7	77.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	76.2	86.5	87.1	85.1	79.2	70.9	63.6	65.9	71.1	77.3	80.7	84.0	(6)	
(7)		DBStd	10.69	6.19	5.89	6.56	7.22	6.49	4.68	5.54	5.95	6.83	7.71	7.42	6.28	(7)
(8)		HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD65	269	0	0	0	2	19	79	93	59	15	2	0	0	(9)
(10)		CDD50	9562	1132	1040	1088	876	648	409	421	493	632	847	921	1055	(10)
(11)		CDD65	4356	667	620	623	427	202	38	50	87	197	384	471	590	(11)
(12)		CDH74	51186	8569	8071	7812	4809	1815	365	512	914	1901	4054	5298	7066	(12)
(13)		CDH80	28479	5222	4943	4599	2478	705	80	115	304	847	2133	2933	4120	(13)
(14)	Wind	WSAvg	5.4	6.2	5.8	5.1	5.0	4.3	4.2	4.3	5.0	5.6	6.2	6.4	6.5	(14)
(15)	Precipitation	PrecAvg	2.4	0.2	0.4	0.4	0.4	0.2	0.2	0.2	0.0	0.1	0.1	0.1	(15)	
(16)		PrecMax	4.3	0.8	1.7	1.5	1.9	0.9	0.5	0.7	0.9	0.4	0.7	0.4	0.4	(16)
(17)		PrecMin	1.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	0.9	0.2	0.4	0.3	0.5	0.2	0.2	0.2	0.2	0.1	0.2	0.1	0.1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	110.5	108.8	107.4	102.8	96.1	86.6	87.8	95.2	98.5	105.3	106.4	108.7	(19)
MCWB			67.2	68.1	67.8	66.1	62.0	58.0	58.2	59.0	61.4	65.1	65.0	67.2	(20)	
2%		DB	105.9	105.2	103.7	98.6	90.9	82.0	83.0	87.0	93.6	99.9	101.9	104.3	(21)	
		MCWB	66.7	67.5	67.4	64.6	59.9	56.1	56.0	56.3	59.1	63.2	63.5	66.2	(22)	
5%		DB	102.3	102.2	100.4	95.3	87.2	78.1	79.4	82.7	89.5	95.9	98.2	100.4	(23)	
		MCWB	66.7	66.8	66.4	63.4	59.4	55.1	54.6	55.2	57.7	61.9	62.6	65.2	(24)	
10%		DB	98.8	99.0	97.1	91.8	83.6	74.6	76.1	79.0	85.0	91.6	94.4	96.7	(25)	
		MCWB	66.0	66.6	65.5	62.6	58.5	54.1	53.6	54.0	56.4	60.7	61.4	64.3	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	77.3	76.5	77.9	69.8	69.2	60.3	59.2	60.2	63.2	71.8	70.1	76.7	(27)
MCDWB			91.8	89.1	96.2	86.5	81.0	78.8	83.7	86.8	92.9	93.3	92.0	94.6	(28)	
2%		WB	73.5	73.5	73.8	68.1	65.4	58.3	57.2	58.4	60.8	67.8	66.4	71.9	(29)	
		MCDWB	90.7	88.8	88.0	89.4	78.2	74.5	78.8	81.7	89.9	90.4	90.1	90.8	(30)	
5%		WB	70.5	71.9	71.2	66.7	62.6	56.9	55.8	56.7	59.2	64.7	64.4	68.9	(31)	
		MCDWB	93.6	91.2	91.1	87.8	79.8	72.6	75.1	77.5	85.3	89.0	90.8	91.5	(32)	
10%		WB	68.4	70.2	68.7	65.2	60.3	55.7	54.8	55.3	57.6	62.5	62.9	66.3	(33)	
		MCDWB	93.6	91.2	92.2	87.0	78.4	70.6	72.9	75.2	81.6	86.9	88.7	92.0	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	23.8	22.9	22.3	20.5	20.8	20.8	22.0	21.9	21.7	22.9	23.4	24.5	(35)
MCDBR			26.8	25.1	24.6	23.9	25.5	26.3	26.9	27.9	28.0	27.8	27.5	27.8	(36)	
MCWBR			7.1	7.3	7.4	8.0	8.9	10.0	10.7	11.0	10.2	9.4	8.2	7.4	(37)	
5% WB		MCDWB	21.2	19.9	20.9	20.0	21.4	21.0	23.8	25.0	25.4	25.8	26.5	25.4	(38)	
		MCWBR	6.9	7.5	7.0	7.4	8.5	8.8	9.9	9.8	9.0	9.2	8.1	8.1	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.350	0.364	0.349	0.324	0.300	0.289	0.295	0.322	0.347	0.356	0.330	0.334	(40)	
taud		2.459	2.444	2.470	2.528	2.559	2.550	2.520	2.422	2.329	2.335	2.462	2.474	(41)		
Ebn at Noon		314	304	299	292	285	281	284	287	294	303	318	320	(42)		
Edh at Noon		38	37	35	30	26	24	26	32	39	41	38	38	(43)		
(44)	All-Sky Solar Radiation	RadAvg	2719	2473	2124	1674	1318	1144	1231	1515	1922	2343	2656	2817	(44)	
(45)		RadStd	93	82	61	56	46	36	41	55	68	62	76	67	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (5 neighbors)	+0.79	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page